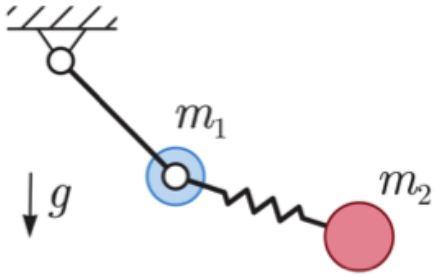
HW2: Lagrangian Dynamics - Springy Double Pendulum

Anas Khmais Hamrouni

Problem setup

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m1 = 0.5      # [kg] - mass in the first joint
m2 = 1.0      # [kg] - mass on the end of spring
l1 = 1.0      # [m] - the length of second link
l2 = 0.8      # [m] - the length of spring at the rest
k = 200       # [N/m] the stifness of the spring
g = 9.81      # [m/s^2] the gravitational acceleration
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We model a planar system with:

- a massless rigid first link of length l_1 attached to a fixed pivot at the origin,
- a point mass m_1 at the end of the first link,
- a massless spring (stiffness k, rest length l_2) attached to m_1,
- a point mass m_2 at the other end of the spring.

We choose generalized coordinates

$$q = \begin{bmatrix} q_1 \\ q_2 \\ \ell \end{bmatrix},$$

where:

- q_1 is the absolute angle of the first link (from the inertial x axis),
- q_2 is the absolute angle of the spring direction (from the inertial x axis),
- ℓ is the instantaneous spring length.

Gravity is $\mathbf{g} = [0, -g]^T$ (downward).

Kinematics

Position of m_1:

$$\mathbf{r}_1 = \begin{bmatrix} l_1 \cos q_1 \\ l_1 \sin q_1 \end{bmatrix}.$$

Position of m_2:

$$\mathbf{r}_2 = \mathbf{r}_1 + \ell \begin{bmatrix} \cos q_2 \\ \sin q_2 \end{bmatrix}.$$

Velocities are $\dot{\mathbf{r}}_i = \frac{d}{dt}\mathbf{r}_i$:

$$\dot{\mathbf{r}}_1 = \begin{bmatrix} -l_1 \sin q_1 \, \dot{q}_1 \\ l_1 \cos q_1 \, \dot{q}_1 \end{bmatrix},$$

$$\dot{\mathbf{r}}_2 = \dot{\mathbf{r}}_1 + \begin{bmatrix} \dot{\ell} \cos q_2 - \ell \sin q_2 \, \dot{q}_2 \\ \dot{\ell} \sin q_2 + \ell \cos q_2 \, \dot{q}_2 \end{bmatrix}.$$

Energy

Kinetic energy:

$$T = \frac{1}{2} m_1 \dot{\mathbf{r}}_1^T \dot{\mathbf{r}}_1 + \frac{1}{2} m_2 \dot{\mathbf{r}}_2^T \dot{\mathbf{r}}_2.$$

Potential energy (gravity + spring):

$$V = m_1 g \, y_1 + m_2 g \, y_2 + \frac{1}{2} k (\ell - l_2)^2,$$

where y_1 and y_2 are the second components of $\mathbf{r}_1$ and $\mathbf{r}_2$.

Lagrangian:

$$\mathcal{L}(q, \dot{q}) = T - V.$$

Euler–Lagrange equations

For $i \in \{1, 2, 3\}$,

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i.$$

Collecting the equations yields the standard manipulator form:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = Q.$$

We use

$$Q = \begin{bmatrix} \tau_1(t, q_1, \dot{q}_1) \\ 0 \\ 0 \end{bmatrix},$$

with:

- Task 2: $\tau_1 = 0$,
- Task 3: $\tau_1(t) = 10(\sin t - q_1) - 0.2 \, \dot{q}_1$.

M, C and g

$M(q)$ the mass matrix, from T:

$$T = \frac{1}{2}\dot{q}^T M(q)\dot{q},$$

and the Coriolis/centrifugal terms via Christoffel symbols:

$$C_{ij}(q,\dot{q}) = \sum_{k=1}^{3} \Gamma_{ijk}(q)\,\dot{q}_k, \quad \Gamma_{ijk} = \frac{1}{2}\left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i}\right),$$

and gravity/spring generalized forces from the potential:

$$g(q) = \nabla_q V(q).$$

Initial condition used in simulation

The assignment specifies the initial cartesian position of the second mass:

$$\mathbf{r}_2(0) = \begin{bmatrix} 1.8 \\ 0 \end{bmatrix}, \quad \dot{q}(0) = 0.$$

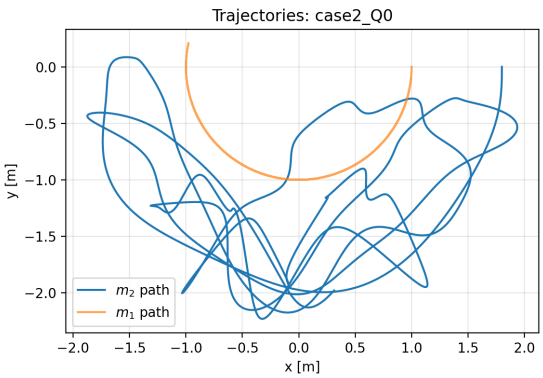
With the coordinate choice above, one rest configuration is:

$$q_1(0) = 0, \quad q_2(0) = 0, \quad \ell(0) = l_2,$$

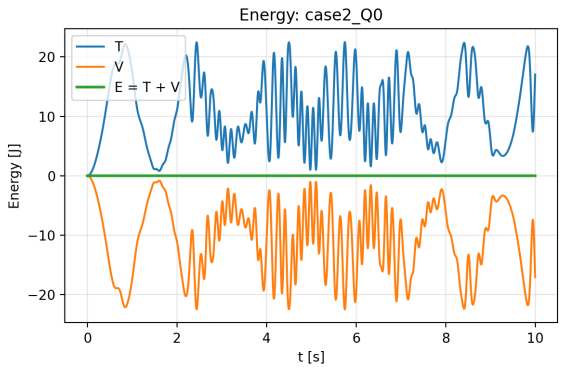
which gives $\mathbf{r}_1(0) = [l_1, 0]^T$ and $\mathbf{r}_2(0) = [l_1 + l_2, 0]^T = [1.8, 0]^T$ for $l_1 = 1.0$ and $l_2 = 0.8$.

Animations

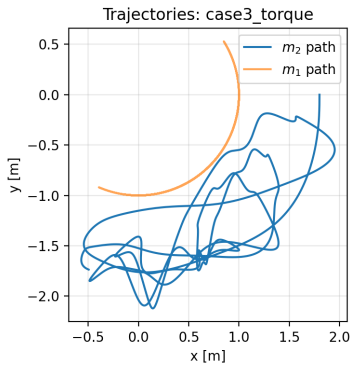
Simulation results (plots)



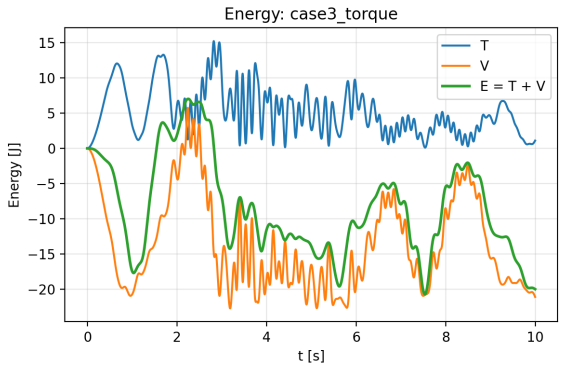
(a) Task 2: trajectories ($Q = 0$).



(b) Task 2: energies ($Q = 0$).



(a) Task 3: trajectories (torque on joint 1).



(b) Task 3: energies (torque on joint 1).